SmartCity ID

Secure Identity Framework for Smart Cities

📋 Project Overview

- A browser-based Role-Based Access Control (RBAC) application
- Two user types: Citizens (read-only) and Admins (full control)
- Single HTML file — no server, no internet required
- Works on desktop, tablet, and mobile devices
- Built with HTML5, CSS3, and Vanilla JavaScript

🎯 Problem Statement

- Smart cities need secure identity management systems
- Citizens need access to their personal records
- Administrators need oversight and control capabilities
- Systems must work offline and be portable
- Must be easy to understand and maintain

✨ Key Features

- 🔐 Secure login with hashed password storage
- 👥 Role-based dashboards (different views for citizens vs admins)
- 📱 Fully responsive design (mobile, tablet, desktop)
- ⚡ Real-time status updates (auto-refresh)
- 📊 Live audit logs of all activities
- 🎨 Futuristic dark theme with animations

System Architecture

- Single Page Application (SPA) with 4 distinct views
- APP Module: Handles routing, events, and UI updates
- BACKEND Module: Encapsulates all business logic & data
- DB Object: Private in-memory database (no persistence)
- Separation of concerns ensures security and maintainability

🖼️ User Interface — Four Pages

- 1️⃣ Login: Authentication entry point for all users
- 2️⃣ Register: New citizen account creation
- 3️⃣ Citizen Dashboard: Personal profile + city status (read-only)
- 4️⃣ Admin Control Room: System overview + logs + users

🔒 Security Design

- ✔ Passwords never stored as plain text — hashed
- ✔ Role-Based Access Control (RBAC) — enforced at 2 layers
- ✔ No persistent storage — session clears when tab is closed
- ✔ IIFE encapsulation — private database inaccessible from console
- ✔ Audit logging — every action tracked with timestamp
- ✔ Input validation — all form fields validated twice

💻 Technology Stack

- Presentation: HTML5 + CSS3 (page structure + dark theme)
- Application Logic: Vanilla JavaScript - APP module
- Business Logic: Vanilla JavaScript - BACKEND module
- Storage: In-Memory JS Object (no external database)
- Fonts/Assets: System Fonts (offline-only, zero dependencies)

✅ Key Advantages

- 🚀 Zero installation — just open HTML file
- 📴 Works offline on any device with browser
- 🔒 Secure by design — no backend vulnerabilities
- 🎓 Educational value — teaches IAM principles
- 🎨 Futuristic UI — engaging animations
- 📱 Fully responsive on all screen sizes

🎯 Use Cases and future Enhnace

- Smart City Government Portal — citizen services
- Educational Prototype — teach IAM concepts
- Demo Platform — showcase without setup
- Training Material — hands-on learning

🚀 Future Enhancements

- Connect to real backend API
- Migrate to secure database
- Upgrade password hashing (bcrypt/Argon2)
- Add email verification
- Implement session timeout
- Add two-factor authentication

SmartCity ID- Secure Identity Framework

A Complete Identity & Access Management System

Thank you